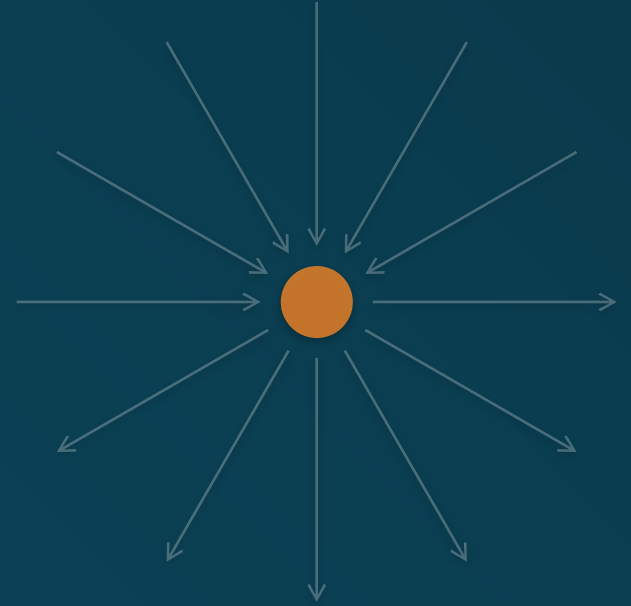


TOPIC 22

Quantum Physics

量子物理



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

22.1 Energy and momentum of a photon 光子的能量与动量

- Photon: quantum of electromagnetic energy, $E = hf$ 光子能量
- The electronvolt; convert eV J 电子伏特换算
- Photon momentum $p = E/c = h/\lambda$ 光子动量

22.2 Photoelectric effect 光电效应

- Threshold frequency and work function 极限频率与逸出功
- Use $hf = \Phi + \frac{1}{2}mv^2_{\max}$ 爱因斯坦光电方程
- Explain why $KE_{\max}$ depends on frequency, not intensity 解释实验规律

22.3–22.4 Wave-particle duality & energy levels 波粒二象性与能级

- de Broglie wavelength $\lambda = h/p$ 德布罗意波长
- Discrete energy levels; $hf = E_2 - E_1$; line spectra 能级跃迁与线光谱

The Photon

光子

Electromagnetic radiation is emitted and absorbed in discrete packets called **photons**. Each photon is a **quantum** of electromagnetic energy.

电磁辐射以不连续的能量包（光子）形式发射和吸收。

$$E = h f = \frac{h c}{\lambda}, h = 6.63 \times 10^{-34} \text{ J s} \quad \text{普朗克常量}$$

- Higher frequency $\Rightarrow$ more energetic photons 频率越高光子能量越大
- Gamma-ray photons $\approx 10^5$ eV; visible light $\approx 2\text{--}3$ eV; radio $\approx 10^{-8}$ eV 量级对比
- Intensity of a beam = rate of arrival of photons $\times$ energy of each photon 光强 = 光子数 $\times$ 单光子能量

The Electronvolt

电子伏特

One **electronvolt (eV)** is the kinetic energy gained by an electron when it is accelerated through a potential difference of one volt.

电子经 1 V 电势差加速所获得的动能。

$$1\text{ eV} = 1.60 \times 10^{-19} \text{ J}; 1\text{ MeV} = 1.60 \times 10^{-13} \text{ J} \quad \text{keV} = 10^3 \text{ eV} \cdot \text{MeV} = 10^6 \text{ eV}$$

- Convenient for photon energies and atomic energy levels 光子能量与原子能级的常用单位
- Convert before substituting into SI equations: $E(\text{J}) = E(\text{eV}) \times 1.60 \times 10^{-19}$ 代入公式先换成 J
- Nuclear energies are usually quoted in MeV 核能量常用 MeV

Momentum of a Photon

光子的动量

Although photons have zero rest mass, they carry **momentum**. This follows from combining $E = hf$ with the wave relation $c = f\lambda$ and relativistic energy $E = pc$.

光子虽无静止质量，却具有动量。

$$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$$

动量与波长成反比

- Shorter wavelength $\Rightarrow$ larger momentum 波长越短动量越大
- Photon momentum explains radiation pressure 解释光压
- Unit: kg m s^{-1} or N s 单位换算注意

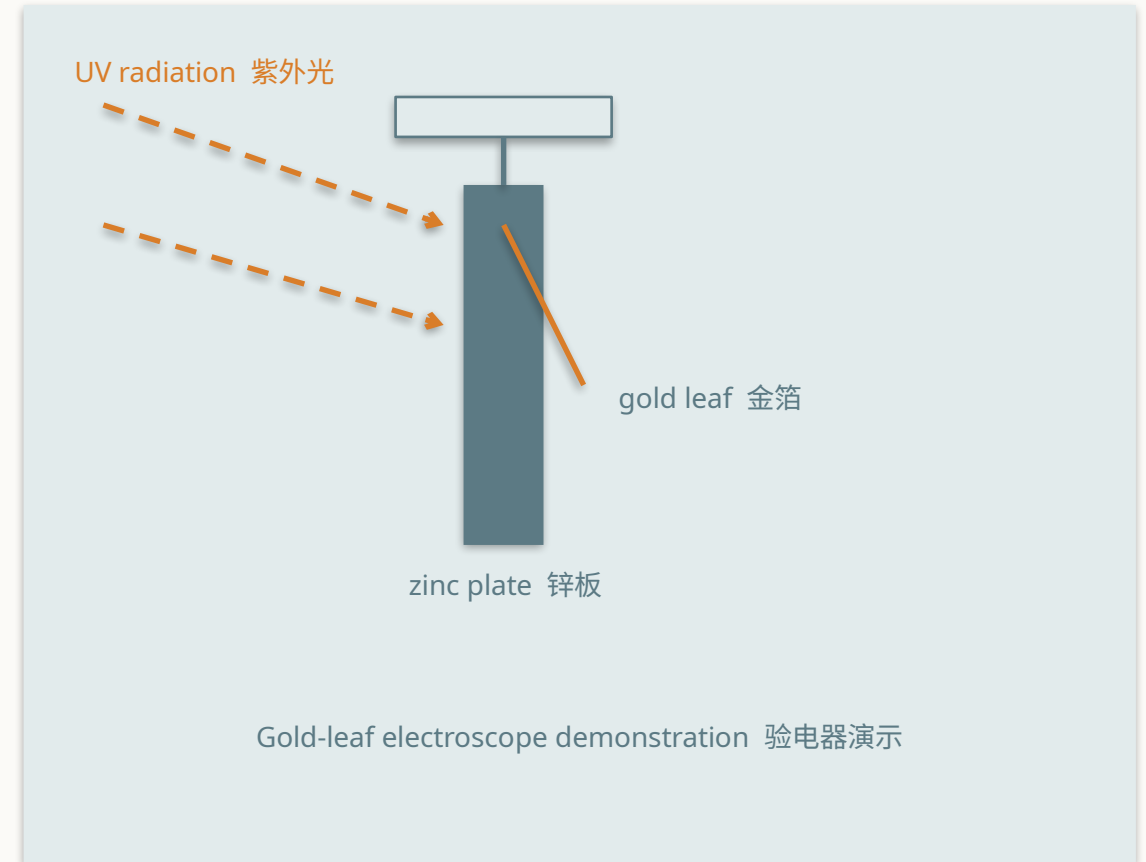
Photoelectric Effect: Observations

光电效应的实验现象

Photoelectric emission is the release of electrons (**photoelectrons**) from the surface of a metal when electromagnetic radiation is incident on it.

电磁辐射照射金属表面使其释放电子的现象。

- UV discharges a **negatively charged** zinc plate instantly; visible light never does 紫外光立即使锌板放电
- Emission is **instantaneous** — no time delay even at low intensity 发射无延迟
- Emission only if $f > \text{threshold frequency } f_0$ (different for each metal) 存在极限频率
- Whether emission occurs depends on **frequency, not intensity** 取决于频率而非光强
- Classical wave theory cannot explain these facts 经典波动理论失效



Einstein's Photoelectric Equation

爱因斯坦光电方程

The **work function** Φ is the minimum energy needed to remove an electron from the surface of a metal. One photon gives all its energy hf to one electron.

逸出功：使电子脱离金属表面所需的最小能量；一个光子对一个电子。

$$h f = \Phi + \frac{1}{2} m v_{max}^2$$

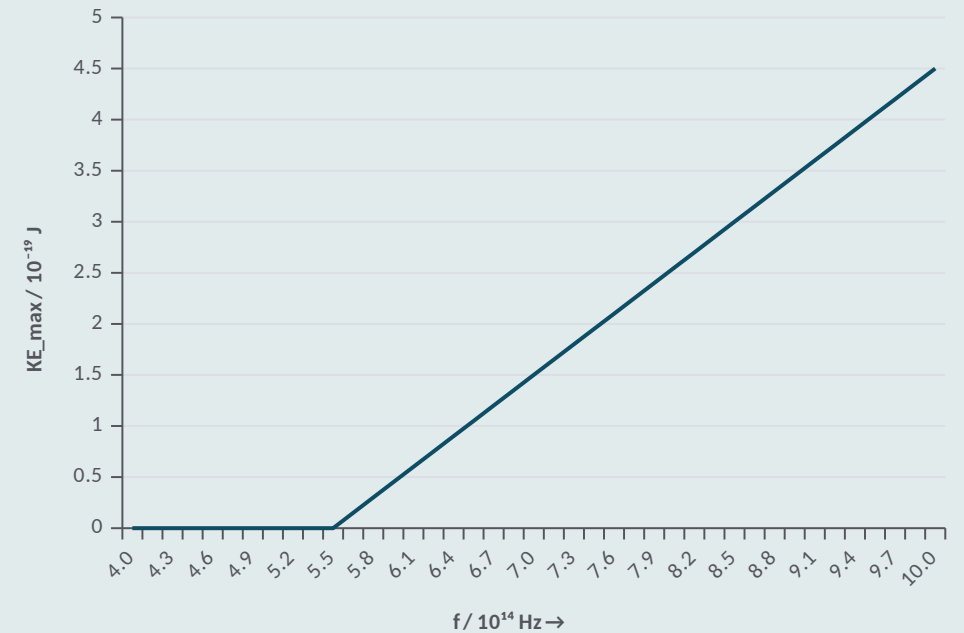
能量守恒：光子能量 = 逸出功 + 最大动能

- Threshold frequency: , i.e. 极限频率
- Any excess photon energy becomes the electron's kinetic energy 多余能量变为动能
- One-to-one photon-electron interaction explains the instant emission 一对一解释瞬时性

Intensity vs Frequency

光强与频率的作用

- **KE_max depends only on frequency:** larger $f \Rightarrow$ more energetic photons 最大动能只随频率变
- **Intensity $\propto$ number of photons:** larger intensity $\Rightarrow$ more electrons per second (photoelectric current) 光强决定光电流
- Graph of KE_max against f is a straight line: gradient = h , intercept = $-\Phi$ 直线斜率 = h
- At constant intensity, higher f means *fewer* photons arrive per second 高光频少光子



KE_max-f graph: gradient h , x-intercept f_0 斜率为 h , 横截距 f_0

Wave-Particle Duality

波粒二象性

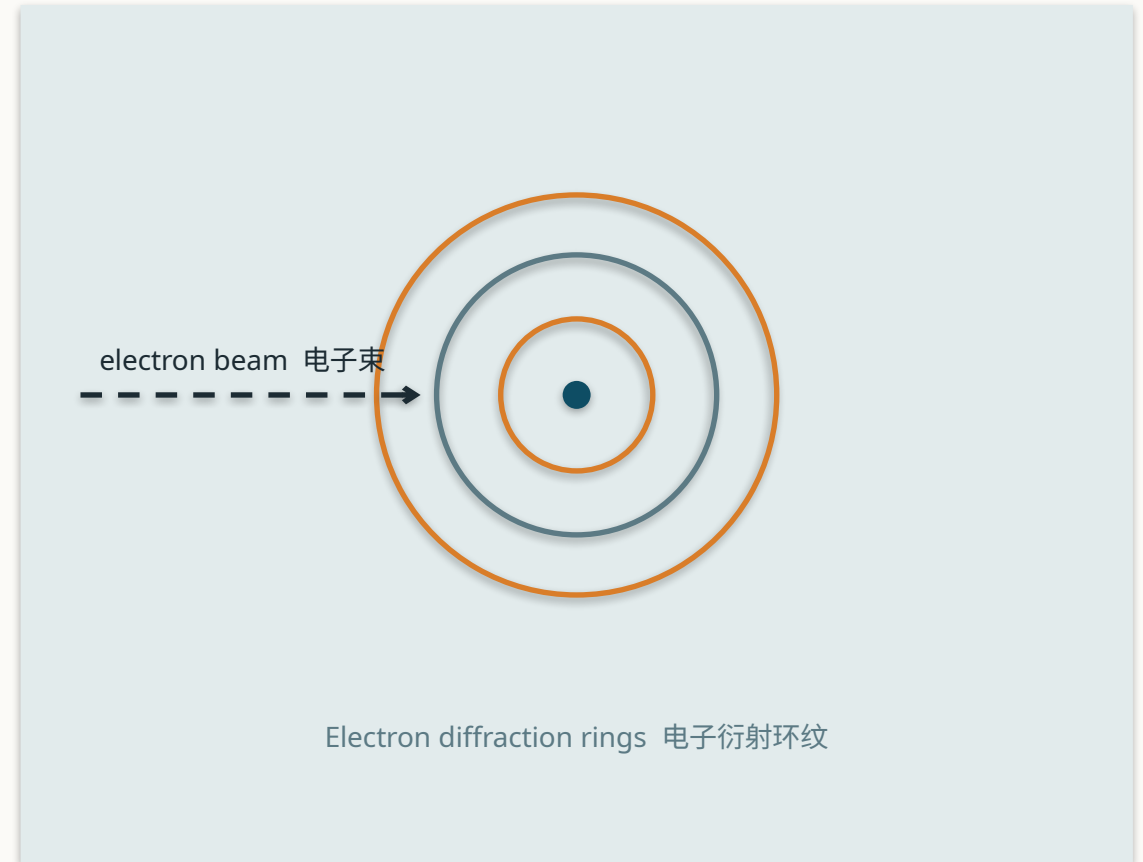
Light shows both wave behaviour (interference, diffraction) and particle behaviour (photoelectric effect).

De Broglie proposed that **all moving particles also have wave properties.**

光具有波粒二象性；德布罗意提出运动粒子也具有波动性。

$$\lambda = \frac{h}{p} = \frac{h}{m v} \quad \text{德布罗意波长}$$

- Electron diffraction rings through thin metal foil confirm the prediction 电子衍射实验证实
- Diffraction occurs when gap size $\approx$ wavelength 波长与缝隙相当时衍射明显
- Everyday objects: λ too small to observe (large m) 宏观物体波长太小



Energy Levels in Atoms

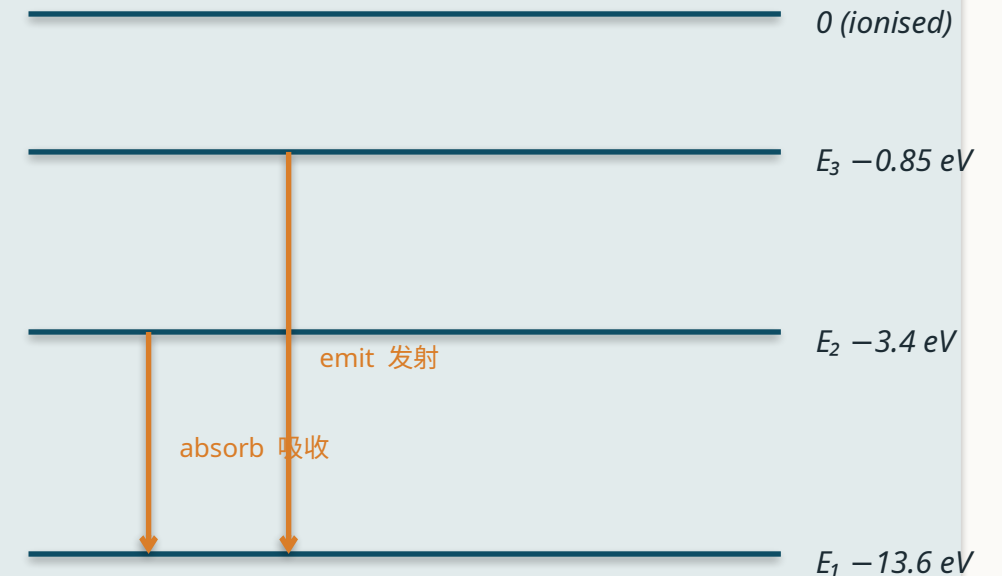
原子的能级

The electron in an atom can occupy only certain **discrete energy levels**. It moves between levels by absorbing or emitting a photon of exactly the energy difference.

原子中的电子只能处于一系列分立的能级，跃迁时吸收或发射特定能量的光子。

$$hf = E_2 - E_1 \quad \text{跃迁能量守恒}$$

- Ground state = lowest level; negative energies mean the electron is bound 基态与束缚态
- Absorption: photon energy must match ΔE exactly 吸收需精确匹配
- Discrete levels explain discrete line spectra 分立能级 $\Rightarrow$ 分立谱线



Line Spectra

线状光谱

- **Continuous spectrum:** hot solid/filament — all wavelengths 连续谱
- **Emission line spectrum:** hot low-pressure gas — bright lines at characteristic wavelengths 发射谱：亮线
- **Absorption line spectrum:** white light through cool gas — dark lines at the same wavelengths 吸收谱：暗线
- Each element has a unique "fingerprint" of lines $\Rightarrow$ chemical analysis of stars 元素指纹
- Line wavelength: 由能级差求波长

Worked Example 22A

例题解析

WORKED EXAMPLE 22A

An electromagnetic wave has speed $3.00 \times 10^8 \text{ m s}^{-1}$ and wavelength 550 nm.

For one photon in this wave, calculate: a) the energy in electronvolts; b) the momentum. ($h = 6.63 \times 10^{-34} \text{ J s}$)

先算焦耳再换算 eV；动量直接用 $p = h/\lambda$ 。

SOLUTION 解答

Step 1 Photon energy J

Step 2 Convert: $E = 3.62 \times 10^{-19} / 1.60 \times 10^{-19} = 2.26 \text{ eV}$

Step 3 Momentum kg m s^{-1}

Answer $E = 2.26 \text{ eV}$; $p = 1.21 \times 10^{-27} \text{ kg m s}^{-1}$

Worked Example 22B

例题解析

WORKED EXAMPLE 22B

The work function energy of platinum is $9.0 \times 10^{-19} \text{ J}$. Calculate: a) the threshold frequency; b) the maximum kinetic energy of a photoelectron when radiation of frequency $2.0 \times 10^{15} \text{ Hz}$ is incident. ($h = 6.63 \times 10^{-34} \text{ J s}$)

光电方程各项统一用焦耳。

SOLUTION 解答

Step 1 Threshold: Hz

Step 2 Photon energy J

Step 3 J ($\approx 2.7 \text{ eV}$)

Answer $f_0 = 1.4 \times 10^{15} \text{ Hz}$; $\text{KE}_{\text{max}} = 4.3 \times 10^{-19} \text{ J}$

Worked Examples 22C & 22D

例题解析

WORKED EXAMPLE 22C

Calculate the de Broglie wavelength of an electron travelling at $2.6 \times 10^7 \text{ m s}^{-1}$.

($h = 6.63 \times 10^{-34} \text{ J s}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$)

1. kg m s^{-1}
- 2.

Answer $\lambda = 2.8 \times 10^{-11} \text{ m}$

WORKED EXAMPLE 22D

Calculate the wavelength emitted when the electron in a hydrogen atom falls from $-0.54 \times 10^{-18} \text{ J}$ to $-2.18 \times 10^{-18} \text{ J}$.

($h = 6.63 \times 10^{-34} \text{ J s}$; $c = 3.00 \times 10^8 \text{ m s}^{-1}$)

1. $\Delta E = 2.18 \times 10^{-18} - 0.54 \times 10^{-18} = 1.64 \times 10^{-18} \text{ J}$
- 2.

Answer $\lambda = 1.21 \times 10^{-7} \text{ m} = 121 \text{ nm}$

Class Practice

随堂练习

Q1

A photon has energy 4.3 eV. Calculate its wavelength and its momentum. ($h = 6.63 \times 10^{-34} \text{ J s}$)

Q2

The work function of zinc is 3.6 eV. Calculate the threshold wavelength for zinc.

Q3

An electron in a hydrogen atom falls from -0.85 eV to -3.4 eV . Calculate the wavelength of the emitted radiation.

Answers Q1: $\lambda = 290 \text{ nm}$, $p = 2.3 \times 10^{-27} \text{ kg m s}^{-1}$ Q2: $\lambda_0 = 345 \text{ nm}$ Q3: $\Delta E = 2.55 \text{ eV}$, $\lambda = 487 \text{ nm}$

Summary

本章小结

KEY FORMULAS 核心公式

$$E = hf = hc/\lambda$$

光子能量

$$1\text{ eV} = 1.60 \times 10^{-19}\text{ J}$$

电子伏特

$$p = h/\lambda$$

光子动量

$$hf = \Phi + \frac{1}{2}mv_{\text{max}}^2$$

光电方程

$$f_0 = \Phi/h$$

极限频率

$$\lambda = h/(mv)$$

德布罗意波长

$$hf = E_2 - E_1$$

能级跃迁

$$\lambda = hc/\Delta E$$

谱线波长

KEY IDEAS 核心思想

- Photon model: one photon $\rightleftharpoons$ one electron 一对一模型
- $E_{\text{K max}}$ set by frequency; current set by intensity 频率定动能，光强定电流
- Matter waves confirmed by electron diffraction 电子衍射证实物质波
- Discrete levels $\Rightarrow$ discrete spectral lines 能级分立 $\Rightarrow$ 谱线分立

Exam Tips

备考提示

- Convert eV to joules **before** substituting into $E = hf$ or $\lambda = hc/\Delta E$. 先换单位再代入。
- Keep $c = f\lambda$ and $E = hf$ separate in your working; mixing wavelength and frequency is the most common slip. λ 与 f 换算用 $c=f\lambda$ 。
- 💡 Explaining intensity: intensity $\propto$ photon **rate**, frequency sets energy of **each** photon. 解释题分清光子数与单光子能量。
- Energy levels are negative; $\Delta E = E_{\text{upper}} - E_{\text{lower}}$ gives a positive photon energy. 能级差取正值。